Transfer Learning for Meta-analysis Under Covariate Shift

Anonymized

***Abstract*—Randomized controlled trials (RCTs) are fundamental to evidence-based medicine, yet they often face challenges in representing broader patient populations. Many current methods in generalizability and transportability rely on frequently violated assumptions in practice. This issue is particularly pronounced in therapeutic areas characterized by disconnected networks and covariate shift. We propose a placebo-anchored transport framework that treats outcomes from source trials as proxy labels and placebo outcomes from the target trial as scarce but high-fidelity gold labels. A sparse correction mechanism anchors baseline risk to the target, and the resulting outcome models are embedded in a doubly robust learner with known propensities to ensure orthogonality and consistency. This approach directly adjusts for site-level heterogeneity, and produces patient-level counterfactual predictions. Experiments for ablation testing and robustness checks demonstrate improved calibration and transport performance, extending the utility of trial evidence to disconnected settings and individualized decision-making.**

***Index Terms*—Transfer learning, meta-analysis, causal inference, clinical trial, doubly robust estimation.**

I. Introduction

Randomized controlled trials (RCTs) remain the gold standard for estimating treatment effects [1], and drug approvals [2]. Yet, their external validity or generalizability remains a major challenge,as trial participants often differ from the patient population of interest [3], thus decision-makers often need to know how a therapy would perform in a different target population than the one studied. This challenge arises frequently in oncology and other therapeutic areas where multiple trials are conducted in distinct populations, and where no single trial provides a direct comparison in the population of interest.

Network meta-analysis (NMA) has become the dominant framework for combining this evidence, allowing estimation of relative treatment effects across a connected network of trials [4, 5]. When individual patient data (IPD) are available, IPD-NMA can incorporate treatment–covariate interactions to explore the heterogeneity of the effect and, in principle, estimate treatment effects in specific subpopulations [6]. However, standard IPD-NMA methods rely on two key conditions: Network connectivity – the target trial must be connected to the network through a shared comparator arm (e.g., placebo or standard of care). Shared-comparator comparability – the comparator arms across trials must be exchangeable after adjustment for measured. [7].

In practice, these conditions are often violated. Network connectivity is violated if target trials are disconnected, for example containing only a placebo arm for the treatment of interest. Baseline risks in shared comparators differ across studies due to covariate shift and unmeasured effect modifiers, violating shared-comparator comparability. IPD-NMA typically models relative effects without anchoring to the observed absolute risk in the target population, leaving results vulnerable to bias when proportional hazards do not hold or when hazard ratios exhibit non-collapsibility. Furthermore, standard NMA outputs are network-wide averages; transporting them to a specific target cohort usually requires meta-regression or marginalization, not patient-level counterfactual prediction.

These limitations create a methodological gap: there is no established approach that can (i) estimate treatment effects in a target population with no treated arm, (ii) correct for covariate shift in both measured and residual baseline risk, and (iii) produce patient-level counterfactuals to support individualized or population-specific decision-making.

We propose a placebo-anchored transport framework to address this gap. Our method learns conditional outcome models from source-trial IPD (treatment + placebo), explicitly calibrates baseline risk using the target trial's observed placebo outcomes, and predicts counterfactual treated outcomes for each target-population patient. This approach requires no network connectivity, directly adjusts for covariate shift through anchoring, and yields target-specific marginal effects alongside patient-level predictions.

II. Related Work

A. Current Methods

Beyond the limitations of IPD-NMA discussed previously, a broader set of IPD-based methods has been developed to address cross-trial heterogeneity. These methods can be categorized into four families: classical parametric modeling approaches, machine learning/nonparametric approaches, Deep learning and ensemble methods, and weighting-based approaches.

Classical parametric modeling approaches: these methods use explicit regression models to estimate treatment effects. The main methods include a one-stage IPD meta-analysis [8], a two-stage IPD meta-analysis [9, 8], a meta-analysis of IPD networks [4, 5, 10], and hierarchical multilevel models [11]. These approaches can be extended to include treatment-covariate interactions to identify subgroups of patients with differential treatment effects [12]. These methods offer interpretability and flexibility, but rely heavily on the correct specification of the model and can perform poorly under misspecification [9], limiting their robustness. In addition, they may be computationally intensive [8], require careful handling of missing data [13], and are sensitive to heterogeneity between trials, which can limit robustness and generalizability in practice. Although classical parametric modeling approaches rely on frequentist estimation, Bayesian methods offer an alternative probabilistic framework that retains similar structural models (e.g., hierarchical or network models) while allowing richer uncertainty quantification and integration of prior information [14].

Non-parametric and machine learning (ML) approaches offer flexible alternatives to classical parametric models. This can be done by relaxing the functional form assumptions and allowing the estimation of heterogeneous treatment effects from IPD data. Tree-based methods such as Causal Forests [15], Generalized Random Forests (GRF) [16], and Bayesian Additive Regression Trees (BART) [17, 18], have been adapted to IPD by utilizing the treatment–covariate interactions and accounting for multi-trial structure these methods can give better estimation compared with classical methods. However, these methods trade-off interpretability, data requirements, causal guarantees, and ease of inference compared to classical parametric models.

Deep learning methods, such as causal neural networks [19] and variational autoencoders [20], can model complex high-dimensional covariate structures in counterfactual estimation. Ensemble methods such as the Super Learner, a stacking-based algorithm for optimal prediction [21], and Targeted Maximum Likelihood Estimation (TMLE), a semi-parametric approach for estimating causal effects, offer better flexibility and double robustness when combined [22]. TMLE can also be integrated with multiple machine learning models to estimate nuisance parameters (e.g., propensity scores, outcome regressions) while preserving valid inference during the estimation. These methods improve performance and robustness, but they suffer from several issues, including prioritizing prediction over causal identifiability, limiting interpretability, requiring strong causal assumptions, and lacking validity across heterogeneous trials.

Weighting-based approaches align the characteristics of a defined target population by reweighing the IPD, therefore improving the generalizability of treatment effect estimates. The classical method of reweighting includes inverse probability weighting (IPW) [23, 24]. More advanced methods include augmented IPW (AIPW) estimators, ensuring double robustness by combining outcome and treatment models[25], and overlap weights by emphasizing patients with common covariate profiles [26]. Additionally, covariate-balancing propensity scores [27] and stable balancing weights [28] seek to directly optimize covariate balance during weighting. These approaches has been formalized in Pearl and Bareinboim's transportability framework, by using causal diagrams and selection nodes to test whether causal effects can be transported across populations and sufficient criteria for transportability [29, 30].

While these weighting methods enhance transportability, they face several limitations. Firstly, all methods assume that all relevant effect modifiers are observed (no unmeasured confounding), which is often unrealistic in practice [31]. Secondly, when propensity scores approach 0 or 1, sensitivity to extreme weights arises leading to unstable estimates with high variance [32]. Thirdly, the methods require positivity/overlap such that every patient type in the target population must have some probability of appearing in the trial, which often fails in practice [32]. Finally, model misspecification of the propensity score can lead to biased estimates, and while doubly robust methods like AIPW can protect against some misspecification, they still fail when both the outcome and propensity score models are wrong. While these methods formalize external validity, they depend on strong, often untestable assumptions, such as conditional exchangeability, positivity, and correct model specification, which can be difficult to verify in practice.Although previously discussed approaches improve conventional IPD-NMA by modeling effect heterogeneity, reweighting, or prediction, a general limitation still exists. That is, they do not directly anchor estimates to observed placebo outcomes in the target trial when providing patient-level counterfactuals in disconnected settings. This motivates the need for alternative frameworks, such as the placebo-anchored transport approach we propose.

B. Our Contributions

This paper develops a calibrated framework for transporting individualized treatment effect estimates across heterogeneous randomized controlled trials under covariate shift. We formulate the transport problem as a *proxy–gold* learning task in the sense of high-dimensional transfer learning, where outcomes from source trials provide abundant but potentially miscalibrated proxy signals, and placebo outcomes observed at the target site provide a limited but informative calibration signal for baseline risk. This formulation does not claim causal identification of target treatment effects from target data alone; instead, it defines a stable working-model estimand suitable for disconnected or weakly connected trial settings.

Within this framework, we introduce a placebo-anchored correction mechanism that adjusts proxy outcome regressions using a restricted, low-complexity function class. The resulting anchoring step explicitly targets systematic baseline miscalibration induced by population heterogeneity and yields calibrated outcome predictions in the target population. The low-complexity restriction is imposed as a regularity condition rather than an identification assumption, allowing estimation error to be decomposed into stochastic error and an explicit structural approximation term.

We further embed the anchoring procedure in a doubly robust learner with cross-fitting, exploiting the fact that treatment assignment is randomized and propensities are known by design in randomized trials. The resulting estimator is Neyman-orthogonal and admits a $\sqrt{N}$ asymptotic linear expansion for the working-model conditional average treatment effect, where N denotes the total sample size across sites. The analysis clarifies that while the leading stochastic term is governed by

N, finite-sample performance depends jointly on the proxy sample size, the number of target placebo observations used for calibration, and approximation error arising from violations of the working model.

Finally, through synthetic and semi-synthetic experiments, including ablations and robustness checks, we empirically characterize the operating regimes of placebo anchoring and illustrate how performance degrades gracefully toward proxy-only baselines as structural assumptions are weakened.

C. Organization of this Paper

The remainder of the paper proceeds as follows. § III formalizes the problem setting, data structure, and identification assumptions. § IV presents the placebo-anchored proxy–gold estimator and the DR-Learner integration in detail. § V evaluates the approach on synthetic ablations, along with robustness checks when some identification assumptions are violated. Finally, § VI summarizes findings, discusses limitations, and outlines directions for future work.

III. Preliminaries

Here we introduce our preliminaries and notation for this paper.

A. Problem Statement

We study the setting where a therapy is evaluated across multiple RCTs, each conducted at a different site or in a different patient population. Within every site, both treatment and placebo arms are observed under a harmonized protocol (identical dosage, randomization scheme, and eligibility criteria). Consequently, treatment assignment is randomized and the propensity model is known by design.

The methodological challenge arises because the covariate distribution of enrolled patients drifts across sites, reflecting population heterogeneity (e.g., the same oncology drug trialed in the U.S. versus Norway). Such covariate shift induces systematic differences in baseline risk and treatment–covariate interactions, leading to biased estimates if outcomes are naively pooled or if network connectivity assumptions are imposed.

Our goal is to estimate patient-level counterfactual outcomes and treatment effects in a target site by leveraging information from source trials while anchoring estimates to the observed placebo outcomes in the target. Following [33]'s proxy–gold paradigm, we treat source-site outcomes as `proxy` labels and the target-site placebo outcomes as scarce but high-fidelity `gold` labels. This enables calibrated transport of treatment effects across heterogeneous populations without requiring network connectivity.

B. Data and Proxy–Gold Setup

We observe multiple randomized controlled trials (RCTs) indexed by $c \in \{1, 2, \ldots, C\}$ (`sources`) together with one `target` trial, indexed by $c = 0$. In each trial c, let $X \in \mathbb{R}^p$ denote baseline covariates, $A \in \{0, 1\}$ the randomized treatment assignment, and Y the outcome. Because treatment is randomized within each trial, the propensity $e_c(x) = \mathbb{P}(A = 1 \mid X = x, c)$ is known by design.

Proxy data (abundant). From the source trials we have individual participant data (IPD) for both treatment and placebo arms,

$$\mathcal{D}_{\text{proxy}} \; = \; \bigcup_{c=1}^{C} \{(X_i^{(c)}, A_i^{(c)}, Y_i^{(c)})\}_{i=1}^{n_c}.$$

These data allow us to fit rich conditional outcome models. However, due to population heterogeneity and covariate shift across sites, such models may be biased when applied directly to the target trial.

Gold data (scarce but high-fidelity). In the target trial $c = 0$, both treatment and placebo arms exist, but the placebo outcomes play a special role: they directly reveal the target population's baseline risk distribution. We denote

$$\mathcal{D}_{\text{gold}} \; = \; \{(X_j^{(0)}, A_j^{(0)} = 0, Y_j^{(0)})\}_{j=1}^{m}.$$

$\mathcal{D}_{\text{proxy}}$ provides a low-variance but potentially biased signal, whereas $\mathcal{D}_{\text{gold}}$ provides a scarce but unbiased calibration signal. We therefore cast the target placebo outcomes as "gold" labels that anchor and debias outcome models trained on the source trials.

C. Identification and Regularity Assumptions

We distinguish carefully between *design-based identification* and *model-based regularity conditions*. Our approach does *not* assume classical cross-site exchangeability, nor does it claim nonparametric identification of the true target-site treatment effect. Instead, we define and estimate a *working-model CATE* whose interpretation and stability rely on explicit structural regularity conditions.

A1 **Consistency.** For each individual,

$$Y = Y(1)\,\mathbb{I}(A = 1) + Y(0)\,\mathbb{I}(A = 0).$$

A2 **Randomization within sites.** Within each trial c, treatment assignment is randomized:

$$(Y(0), Y(1)) \perp A \mid X,\, c.$$

Consequently, the propensity score $e_c(x) = \mathbb{P}(A = 1 \mid X = x, c)$ is known by design.

A3 **Positivity (overlap).** There exists $\epsilon > 0$ such that for all sites c and all x in the support,

$$\epsilon \le e_c(x) \le 1 - \epsilon.$$

This is a design property of randomized controlled trials and applies only to baseline covariates measured prior to treatment assignment.

A4 **Outcome-regression stability (regularity).** For each arm $a \in \{0, 1\}$ and site c, the conditional outcome regression

$$\mu_{a,c}(x) := \mathbb{E}[Y \mid A = a, X = x, c]$$

is Lipschitz continuous in x with a site-uniform constant $L < \infty$:

$$\big|\mu_{a,c}(x) - \mu_{a,c}(x')\big| \le L\|x - x'\|, \quad \forall x, x'.$$

This assumption is a *regularity condition* ensuring that moderate covariate shift does not induce arbitrarily large changes in predicted outcomes. It does *not* imply exchangeability or transportability across sites.

A5 Low-complexity transport bias (approximation condition). Differences between site-specific outcome regressions and proxy regressions are structured and of low complexity. Specifically, for each arm a and site c,

$$\mu_{a,c}(x) = \mu_a^{\mathrm{proxy}}(x) + \delta_{a,c}(x),$$

where $\delta_{a,c}$ belongs to a restricted function class $\mathcal{D}$ (e.g., sparse linear functions $\delta(x) = \beta^\top x$ with $\|\beta\|_0 \le s \ll p$). This assumption is *not* an identification assumption. It defines a working approximation class that enables stable estimation and calibrated extrapolation. When A5 holds approximately, estimation error decomposes into stochastic error and approximation error; when violated, performance degrades smoothly toward proxy-only baselines.

A6 Cross-arm coupling for treated-arm extrapolation. When treated outcomes are unavailable at the target site, the treated-arm transport bias is linked to the placebo-arm bias through a structural coupling:

$$\delta_{1,0}(x) = \rho\, \delta_{0,0}(x) + \zeta(x), \qquad \rho \in [0,1],$$

where ζ captures treatment-specific deviations not explained by placebo anchoring.

a) Interpretation.: Assumptions **A1–A3** are standard design-based conditions for randomized trials. Assumptions **A4–A6** are regularity and approximation conditions that define a well-posed working estimand rather than a nonparametrically identified causal effect. Throughout, we emphasize calibrated estimation under acknowledged transport bias rather than causal identification under strong exchangeability assumptions.

IV. Proposed Framework

A central premise of our framework is that every trial is randomized, so the true propensities $e_c(x)$ are known by design. This ensures that any doubly robust estimator built on top of site-specific outcome models is consistent as long as the outcome regressions are well calibrated. The key methodological challenge is therefore not treatment assignment, but *covariate shift across sites*: outcome models trained on sources may be systematically biased when applied to the target population. We address this challenge by treating source trial outcomes as `proxy` labels and target placebo outcomes as scarce but high-fidelity `gold` labels for calibration. This can be seen as a combination of [33]'s multi-task transfer learner for data-fusion and [34]'s Doubly Robust (DR)-Learner

A. Three-Stage Placebo-Anchored Doubly Robust Estimator

a) Stage 1: Proxy fit on source trials.: We first train outcome regressions on pooled source IPD:

$$\widehat{\mu}_a^{\mathrm{proxy}}(x) \;\approx\; \mu_a(x), \quad a \in \{0,1\}, \tag{1}$$

using any flexible learner (linear model, random forest, or neural net). Since n_{proxy} is large, these estimators have low variance, but due to cross-site heterogeneity their predictions may be systematically miscalibrated for the target site.

Remark. Pooling source trials in Stage 1 does not assume outcome homogeneity across sites; it serves only to reduce variance. Systematic cross-site bias is addressed explicitly in Stage 2 via Assumption **A5**.

b) Stage 2: Gold correction on target placebo.: Let the target placebo (gold) sample be

$$\mathcal{D}^{(0)}_{\mathrm{gold},0} = \{(X_j, Y_j) : c_j = 0,\ A_j = 0\}, \qquad m_0 = \big|\mathcal{D}^{(0)}_{\mathrm{gold},0}\big|.$$

Given the proxy baseline predictor $\widehat{\mu}_0^{\mathrm{proxy}}(\cdot)$ from Stage 1, define the residualized gold outcome

$$\widetilde{Y}_j^0 \;:=\; Y_j \;-\; \widehat{\mu}_0^{\mathrm{proxy}}(X_j), \qquad (X_j, Y_j) \in \mathcal{D}^{(0)}_{\mathrm{gold},0}.$$

We estimate a sparse transport-bias correction $\delta_{0,0}(x) = \delta_{0,0}^\top x$ by solving the LASSO regression

$$\widehat{\delta}_{0,0} \;\in\; \arg\min_{\delta \in \mathbb{R}^p} \left\{ \frac{1}{m_0} \sum_{j: c_j = 0, A_j = 0} \Big(\widetilde{Y}_j^0 - \delta^\top X_j \Big)^2 \;+\; \lambda_0 \|\delta\|_1 \right\}. \tag{2}$$

This is equivalent to [33]'s proxy–gold correction after the reparameterization $\delta = \beta - \widehat{\beta}_{\mathrm{proxy}}$, except that here the proxy predictor $\widehat{\mu}_0^{\mathrm{proxy}}(\cdot)$ may be any flexible ML model rather than a linear regression.

The anchored placebo outcome regression is then

$$\widehat{\mu}_{0,0}^{\mathrm{anch}}(x) \;:=\; \widehat{\mu}_0^{\mathrm{proxy}}(x) \;+\; \widehat{\delta}_{0,0}^\top x.$$

We choose λ_0 by cross-validation on $\mathcal{D}^{(0)}_{\mathrm{gold},0}$.

c) Transport to the treated arm.: Below are two identified choices; use Option A if the target contains treated outcomes, and Option B if it does not.

Option A (preferred when target treated outcomes exist): estimate a treated correction using target treated data. If the target trial also contains treated observations, define

$$\mathcal{D}^{(0)}_{\mathrm{gold},1} = \{(X_j, Y_j) : c_j = 0,\ A_j = 1\}, \qquad m_1 = \big|\mathcal{D}^{(0)}_{\mathrm{gold},1}\big|.$$

Residualize the treated gold outcomes using the proxy treated predictor $\widehat{\mu}_1^{\mathrm{proxy}}(\cdot)$:

$$\widetilde{Y}_j^1 \;:=\; Y_j \;-\; \widehat{\mu}_1^{\mathrm{proxy}}(X_j), \qquad (X_j, Y_j) \in \mathcal{D}^{(0)}_{\mathrm{gold},1}.$$

Estimate $\delta_{1,0}(x) = \delta_{1,0}^\top x$ via

$$\widehat{\delta}_{1,0} \;\in\; \arg\min_{\delta \in \mathbb{R}^p} \left\{ \frac{1}{m_1} \sum_{j: c_j = 0, A_j = 1} \Big(\widetilde{Y}_j^1 - \delta^\top X_j \Big)^2 \;+\; \lambda_1 \|\delta\|_1 \right\}. \tag{3}$$

Then define the anchored treated outcome regression

$$\widehat{\mu}_{1,0}^{\mathrm{anch}}(x) \;:=\; \widehat{\mu}_1^{\mathrm{proxy}}(x) \;+\; \widehat{\delta}_{1,0}^\top x.$$

Choose λ_1 by cross-validation on $\mathcal{D}^{(0)}_{\mathrm{gold},1}$.

Option B (when target has no treated arm): impose a cross-arm bias restriction. If the target trial has no treated arm (disconnected setting), then $\delta_{1,0}$ is not identifiable from target data. This choice defines the working-model treated regression $\mu^\star_{1,0}$ (and hence $\tau^\star$) in the disconnected setting; it is not implied by design-based identification. This corresponds to the special case $\rho = 1$ and $\zeta \equiv 0$ in Assumption **A6**, and is adopted as a pragmatic modeling choice in disconnected settings rather than as a consequence of identification.

A simple identified restriction consistent with Assumption **A5** is to posit a shared sparse bias mechanism across arms:

$$\delta_{1,0} = \delta_{0,0}.$$

In this case, define

$$\widehat{\mu}^{\mathrm{anch}}_{1,0}(x) \;:=\; \widehat{\mu}^{\mathrm{proxy}}_{1}(x) \;+\; \widehat{\delta}^{\top}_{0,0}x.$$

d) Joint placebo-anchored CATE.: Finally, define the placebo-anchored CATE estimate in the target population:

$$\widehat{\tau}(x) \;:=\; \widehat{\mu}^{\mathrm{anch}}_{1,0}(x) \;-\; \widehat{\mu}^{\mathrm{anch}}_{0,0}(x).$$

e) Stage 3: Orthogonalization and CATE regression.: Given cross-fitted nuisance estimates, we form the doubly robust pseudo-outcome

$$\psi_i := \widehat{\tau}(X_i) + \frac{A_i - e_{c_i}(X_i)}{e_{c_i}(X_i)\{1 - e_{c_i}(X_i)\}}\Big\{Y_i - \widehat{\mu}^{\mathrm{anch}}_{A_i,c_i}(X_i)\Big\}. \tag{4}$$

We then estimate the working-model CATE function by regressing ψ_i on X_i (e.g., kernel regression, random forest, or a sparse linear model) using cross-fitting:

$$\widehat{\tau}_{\mathrm{DR}}(\cdot) \approx \mathbb{E}[\psi \mid X = \cdot].$$

In particular, under Assumptions A4–A6 and standard DML regularity conditions, $\widehat{\tau}_{\mathrm{DR}}(x)$ admits a $\sqrt{N}$-rate asymptotic linear expansion around the *working-model* target

$$\tau^\star(x) := \mu^\star_{1,0}(x) - \mu_{0,0}(x),$$

where $\mu^\star_{1,0}$ is defined by the anchoring rule (Option A) or by the cross-arm coupling (Option B). We emphasize that $\tau^\star(x)$ is a stable *working estimand*; it need not equal the true target CATE $\tau_0(x)$ when Assumptions A5–A6 are violated.

Since we are working with RCTs, the propensity scores e_{c_i} for each site are known in advance from the protocol, or can be accurately estimated empirically with no unobserved confounders. This ensures that the estimator in (4) is asymptotically linear and admits a $\sqrt{N}$-rate expansion around the working-model CATE $\tau^\star(x)$ under standard DML regularity conditions. [35, 34].

In short, Stage 1 provides a low-variance proxy model from source RCTs; Stage 2 debiases the outcome model using target gold data; and orthogonalization delivers robustness to residual errors. Stage 3 then orthogonalizes the scores to ensure that we have a doubly robust estimator for the CATE.

V. Experiments

We evaluate whether the proposed placebo–anchored proxy–gold DR-Learner (i) improves *target calibration*, (ii) *transport accuracy* under covariate shift, (iii) and yields lower error for patient-level CATEs.

A. Ablation Study

To isolate the contribution of each component in the proposed framework, we conducted an ablation study under the synthetic multi-center RCT setup with covariate shift. We compared four model variants:

(i) `No-Transfer`, which relies solely on placebo outcomes from the target site and cannot extrapolate treated outcomes;
(ii) `Proxy-Only`, which uses pooled source trials without anchoring;
(iii) `Anchor-Only`, which incorporates placebo anchoring but omits doubly robust correction;
(iv) `Proposed`, which integrates proxy learning, placebo anchoring, and doubly robust orthogonalization.

These ablations therefore serve as stress tests of the framework's design principles, quantifying the distinct contributions of anchoring for target calibration and orthogonalization for robust CATE estimation. The source of benefits evaluated by their absence in each ablation is summarized in Table I.

Variant	Omitted / Stress-Tested
`No-Transfer`	No proxy or anchoring; relies only on placebo.
`Proxy-Only`	No anchoring; proxy models under covariate shift.
`Anchor-Only`	No orthogonalization; anchoring without DR.
`Proposed`	Full model with proxy, anchoring, and DR.

TABLE I
Design components isolated by each ablation variant.

B. Synthetic Data Generation

To evaluate the proposed estimator under controlled conditions, we generate multi-center RCT data with explicit covariate shift and known ground truth treatment effects. Each simulated trial corresponds to a distinct site $c \in \{0, 1, \dots, C\}$ with site-specific covariate distributions. We base the following design off a similar study in multi-site RCTs (cf. [36]).

a) Covariates.: For each subject, we sample $p = 5$ baseline covariates $X \in \mathbb{R}^5$. Three covariates are designated as relevant effect modifiers, contributing directly to treatment heterogeneity, while the remaining two are nuisance covariates. To induce site-level heterogeneity, each site c has a shifted Gaussian distribution for X, with site-specific means drawn from $\mathcal{N}(0, 1)$ and variance fixed at 1 per dimension.

b) Treatment assignment.: Treatment is randomized independently of covariates, with probability $P(A = 1) = 0.5$ for each site. This ensures that the true propensity scores are known and equal across all sites.

c) Outcomes.: Continuous outcomes are generated as

$$Y = \mu_0(X) + A \cdot \tau(X) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.5^2),$$

where the baseline response $\mu_0(X)$ is a linear function of all covariates, and the conditional treatment effect $\tau(X)$ depends only on the three relevant covariates. This setup produces structured heterogeneity while maintaining analytic tractability.

d) Target and source sites.: One site ($c = 0$) is designated as the target trial, from which placebo outcomes are used as gold labels. All other sites provide source data, supplying abundant proxy outcomes but subject to covariate shift. By design, the target and sources share overlapping support but differ in their covariate distributions, mimicking real-world population drift across trial locations.

e) Implementation.: The simulation is implemented in Python with fixed random seeds to ensure reproducibility.

C. Results of Ablation Study

Performance was evaluated using three complementary criteria: Precision in Estimation of Heterogeneous Effects (PEHE) i.e., root mean-squared error (RMSE) of the CATE, mean absolute error of the average treatment effect (ATE), and calibration RMSE for treated and placebo outcome regressions (μ_1 and μ_0). These metrics jointly capture transport accuracy at both the individual and population levels. The results are summarized in Table II, and visualized in Figures 1, 2, 3, and 4 where lower values indicate better performance.

Variant	**PEHE**	**MAE of ATE**	**CalRMSE μ_0**	**CalRMSE μ**
`No-Transfer`	0.835	0.618	**0.010**	0.826
`Proxy-Only`	0.182	0.147	0.188	0.214
`Anchor-Only`	0.204	**0.092**	0.014	0.214
`Proposed`	**0.176**	0.139	0.014	**0.191**

TABLE II

ABLATION RESULTS ON SYNTHETIC MULTI-CENTER RCT SIMULATION. LOWER VALUES INDICATE BETTER PERFORMANCE.

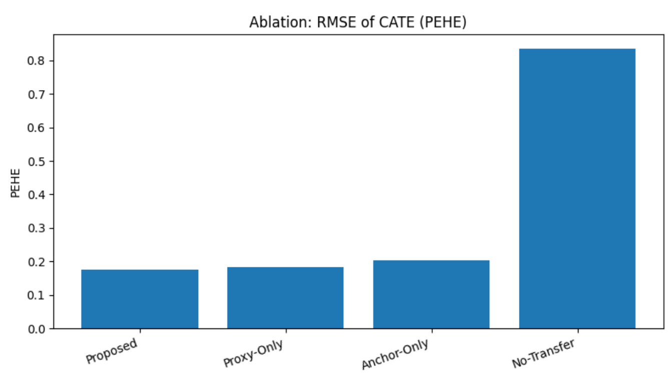

Fig. 1. Bar Plot of PEHE of CATE of ablation study. Lower is better.

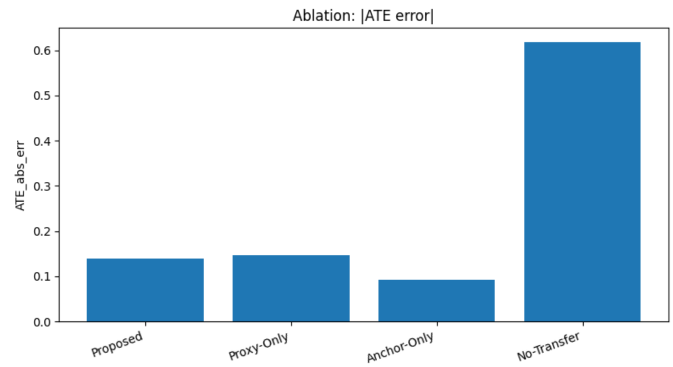

Fig. 2. Bar Plot of MAE of ATE of ablation study. Lower is better.

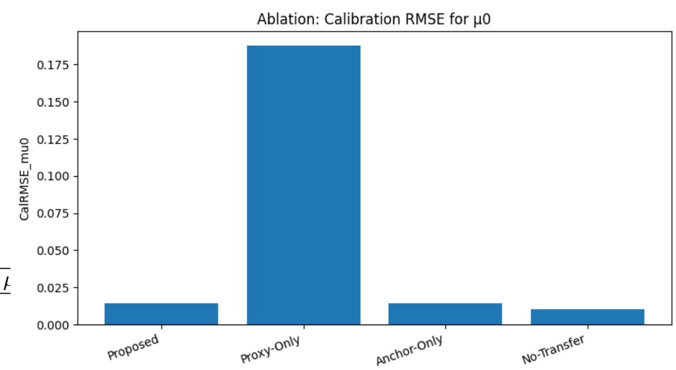

Fig. 3. Bar Plot of Calibration RMSE of placebo arms outcome of ablation study. Lower is better.

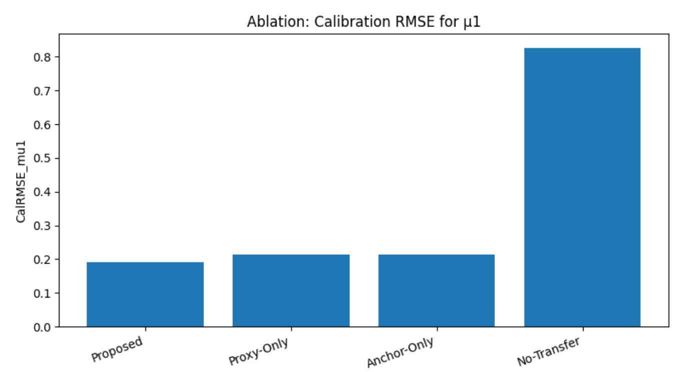

Fig. 4. Bar Plot of Calibration RMSE of treated arms outcome of ablation study. Lower is better.

The `No-Transfer` baseline fails to recover heterogeneity in treatment effects, leading to high PEHE and ATE error despite excellent calibration of placebo outcomes. Incorporating proxy information from source trials substantially reduces error, with the `Proxy-Only` and `Anchor-Only` models both improving CATE estimation but differing in calibration accuracy. Notably, `Anchor-Only` attains the lowest ATE error, as directly anchoring to target-placebo outcomes sharpens marginal effect estimates, though at the cost of higher PEHE due to limited robustness at the patient level. Conversely, `Proxy-Only` inherits bias in baseline calibration from the source trials, resulting in larger ATE error even while improving heterogeneity recovery. By contrast, the full `Proposed` method strikes a favorable balance, achieving the lowest PEHE while also maintaining competitive calibration across both placebo and treated arms. This reflects the complementary roles of anchoring, which debiases baseline risks, and orthogonalization, which stabilizes CATE estimation. Together, these components are necessary to achieve well-calibrated, patient-level transport of trial evidence under covariate shift. Importantly, these empirical patterns are consistent with Assumptions **A4**–**A5**: bounded site-level covariate shift (Assumption **A4**) ensures that anchoring corrects systematic baseline bias, while sparse transport bias (Assumption **A5**) allows orthogonalization to recover fine-grained heterogeneity without overfitting.

D. Robustness Check

We formalize each robustness dimension considered in our experiments.

*a) Shift–Sparsity Sensitivity (Assumptions **A4/A5**).:* Let $\mathbb{P}_{X|c}$ denote the covariate distribution at site c. The total-variation (TV) shift across two sites c_1, c_2 is

$$\Delta(c_1, c_2) \;=\; d_{\mathrm{TV}}\big(\mathbb{P}_{X|c_1},\, \mathbb{P}_{X|c_2}\big) = \tfrac{1}{2}\int |p_{c_1}(x) - p_{c_2}(x)|\, dx.$$

We sweep the bound Δ to vary the degree of population heterogeneity. The transport bias at site c for arm a is modeled as

$$b_{a,c}(x) \;=\; \beta_{a,c}^{\top} x, \quad \|\beta_{a,c}\|_0 \le s,$$

where s controls the sparsity of site-specific deviations.

b) Gold Budget.: Let $m = |\mathcal{D}_{\text{gold}}|$ denote the number of placebo outcomes available in the target. We vary m to assess the efficiency of anchoring and the marginal value of additional "gold" labels.

c) Placebo-to-Treated Transfer Validity.: Bias terms across arms are coupled by

$$\beta_{1,c} \;=\; \rho \cdot \beta_{0,c} + \sqrt{1-\rho^2}\,\eta_c,$$

where $\rho \in [0, 1]$ measures correlation between placebo and treated biases and η_c is independent noise. When $\rho = 1$, bias transfers perfectly; when $\rho = 0$, placebo anchoring is uninformative for treated outcomes.

d) Outcome Misspecification.: We define the outcome regression in nonlinear form:

$$Y = f_a(X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0,$$

where f_a may include interaction terms, splines, or heteroskedastic variance $\mathrm{Var}(\varepsilon \mid X)$. This deviates from the linear specification assumed in Stage 2.

e) Propensity and Overlap Stress.: Within site c, let $e_c(x) = \mathbb{P}(A = 1 \mid X = x, c)$. We vary site-level assignment probabilities and enforce overlap erosion:

$$\exists\, x : \; e_c(x) \approx 0 \;\text{ or }\; e_c(x) \approx 1,$$

to test robustness of the doubly robust learner under positivity violations.

E. Robustness Check Results

We now evaluate the placebo–anchored DR-Learner under a suite of robustness checks that vary covariate shift, sparsity of transport bias, gold sample size, placebo–treated bias transfer, and site propensities. Performance is assessed by PEHE, absolute ATE error, and calibration error for placebo (μ_0) and treated (μ_1) regressions. Representative PEHE curves are shown in Figures 5–9.

*a) Shift–Sparsity Sensitivity (Assumptions **A4/A5**).:* Figure 5 shows that as total-variation shift Δ increases, all methods deteriorate. The `Proposed` model maintains the lowest PEHE under moderate sparsity, benefiting from anchoring and DR correction. As sparsity s increases (Figure 8), anchoring becomes less effective, and performance converges toward `Proxy-Only`, confirming Assumption **A5** as a key driver of robustness.

b) Gold Budget.: Increasing the number of target-placebo labels m (Figure 6) quickly improves calibration, with diminishing returns beyond moderate sample sizes. The `Proposed` model benefits disproportionately, as DR orthogonalization amplifies even modest anchoring, while `Anchor-Only` stabilizes but fails to reach comparable PEHE.

c) Placebo-to-Treated Transfer Validity.: Varying the coupling ρ between placebo and treated bias (Figure 7) reveals that anchoring transfers well when $\rho \ge 0.7$. When ρ decreases, placebo calibration remains intact but treated outcomes degrade, highlighting the necessity of sparse cross-arm bias structure in Assumption **A5**.

d) Propensity and Overlap Stress.: Figure 9 varies site propensities e_c. When overlap is limited, all models degrade, but the `Proposed` estimator consistently outperforms baselines, leveraging known randomization propensities for doubly robust correction. Nevertheless, extreme violations of positivity remain failure modes, consistent with theory.

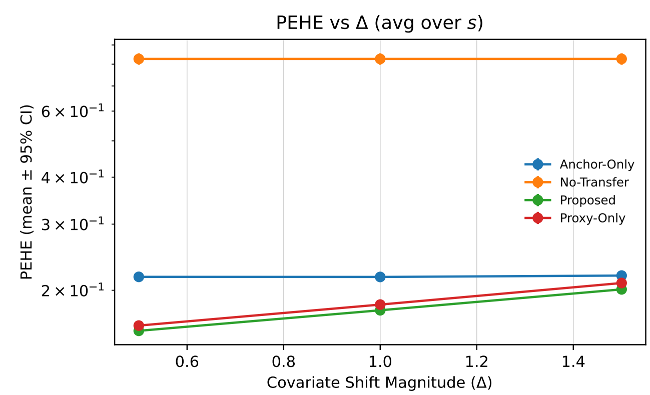

Fig. 5. PEHE vs. total-variation shift Δ (mean ± 95% CI), averaged over sparsity.

VI. Conclusion

We presented a placebo–anchored proxy–gold framework for transporting trial evidence under covariate shift. Source-site IPD provide low-variance proxy signals; target-placebo outcomes serve as scarce but high-fidelity gold labels that calibrate baseline risk via a sparse correction. Embedding these anchored nuisance models in a DR-Learner with known site propensities yields an estimator that is flexible and orthogonal, enabling patient-level counterfactual prediction without requiring network connectivity. Across synthetic ablations and robustness checks, the method improved target calibration and

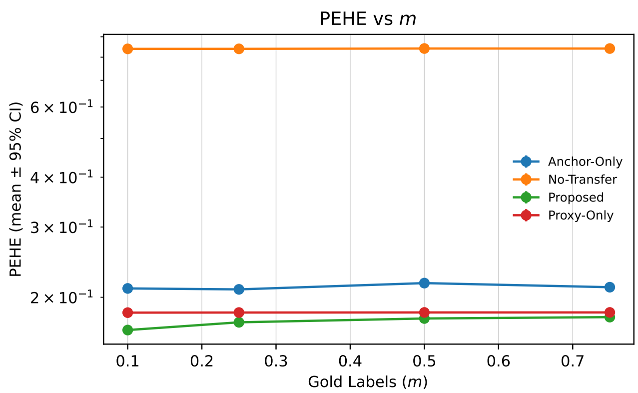

Fig. 6. PEHE vs. number of gold labels m (mean ± 95% CI).

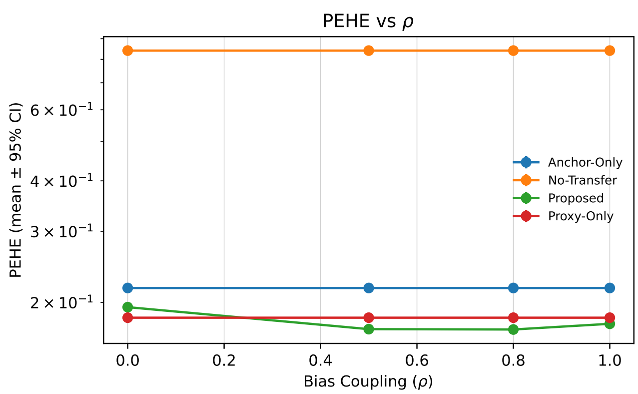

Fig. 7. PEHE vs. cross-arm bias coupling ρ (mean ± 95% CI).

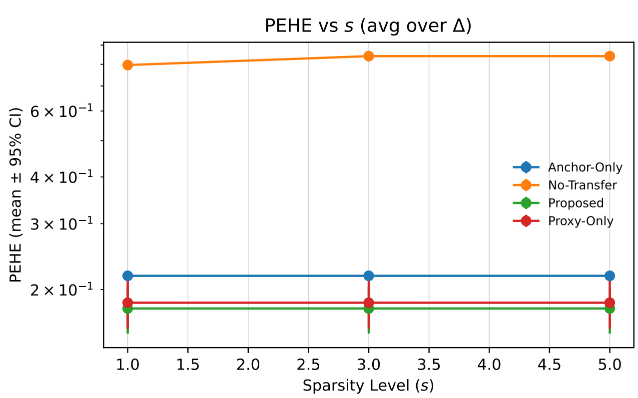

Fig. 8. PEHE vs. sparsity level s (mean ± 95% CI), averaged over Δ.

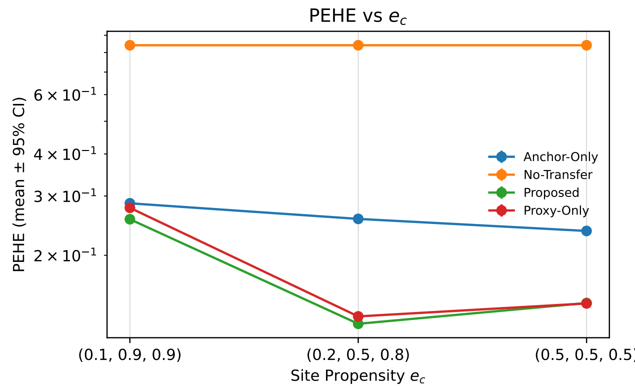

Fig. 9. PEHE vs. site propensity e_c (mean ± 95% CI).

CATE accuracy relative to strong baselines. We also stress-tested the framework against its main assumptions and potential weaknesses. The robustness checks showed consistently promising behavior and performance remained competitive under moderate violations. These findings suggest that placebo anchoring and doubly robust correction offer a practical path for extending the value of trial evidence to settings where conventional network meta-analysis fails. By delivering calibrated, patient-level predictions, our framework supports individualized treatment decisions and improves confidence in applying trial evidence across diverse populations. Most importantly, robustness checks show that these benefits persist even when clinical data deviate from ideal assumptions, underscoring the framework's promise as a tool for evidence translation in heterogeneous and underrepresented populations.

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APPENDIX

ASYMPTOTIC NORMALITY AND FINITE-SAMPLE ERROR DECOMPOSITION FOR THE PLACEBO–ANCHORED CATE ESTIMATOR

This section formalizes two claims:

- the Stage 3 estimator admits an asymptotically linear expansion and a $\sqrt{N}$-CLT for a *working-model* target CATE, and
- the finite-sample error decomposes into a sampling term governed by the total Stage 3 sample size N, a second-order nuisance term governed by both the proxy sample size and the target placebo (gold) sample size, and an explicit structural approximation term governed by Assumptions **A5**–**A6**.

Throughout, we emphasize that Neyman orthogonality provides robustness to nuisance estimation error for the *working-model* estimand, but does not imply semiparametric efficiency for the *true* target CATE $\tau_0(x)$, which is not nonparametrically identified in disconnected regimes.

A. Setup and target parameter

Let $S \in \{0, 1, \dots, C\}$ index sites, with $S = 0$ the target site and $S = c \geq 1$ source sites. For each unit observe $O = (X, A, Y, S)$ with $A \in \{0, 1\}$ and potential outcomes $Y(1), Y(0)$. Define site- and arm-specific outcome regressions

$$\mu_{a,c}(x) := \mathbb{E}[Y \mid A = a, X = x, S = c], \\ a \in \{0, 1\}, \qquad c \in \{0, \dots, C\}.$$

The *true* target-site CATE is

$$\tau_0(x) := \mathbb{E}[Y(1) - Y(0) \mid X = x, S = 0] \\ = \mu_{1,0}(x) - \mu_{0,0}(x).$$

In disconnected settings with no treated outcomes in the target site, $\mu_{1,0}$ (and thus τ_0) is not nonparametrically identified.

a) Working-model target.: We define a *working-model* treated regression $\mu^\star_{1,0}$ via the anchoring/coupling construction described in Section IV. The corresponding working-model target CATE is

$$\tau^\star(x) := \mu^\star_{1,0}(x) - \mu_{0,0}(x),$$

where $\mu_{0,0}$ is identified from target placebo (gold) data on its support, and $\mu^\star_{1,0}$ is the population limit induced by Stage 1 proxy learning, Stage 2 gold calibration, and (if needed) cross-arm coupling. All asymptotic statements in this appendix pertain to $\tau^\star(x)$ (pointwise in x) rather than $\tau_0(x)$.

B. Estimator: Stage 3 orthogonal pseudo-outcome regression

Let n_c be the sample size in site c and let

$$N := \sum_{c=0}^{C} n_c, \qquad \pi_c := \frac{n_c}{N}.$$

Within each site c, treatment is randomized with known propensity $e_c(x) = \mathbb{P}(A = 1 \mid X = x, S = c)$. By Assumption **A3** there exists $\epsilon > 0$ such that $\epsilon \leq e_c(x) \leq 1 - \epsilon$ on the relevant support.

a) Cross-fitted anchored outcome regressions.: Let $\widehat{\mu}^{\text{anch}}_{a,c}$ denote the outcome regression used inside the DR score for site c: for the target site $c = 0$, $\widehat{\mu}^{\text{anch}}_{a,0}$ is the anchored regression from Stage 2 (Option A or B); for source sites $c \geq 1$, $\widehat{\mu}^{\text{anch}}_{a,c}$ may be taken as the site-specific regression fit in Stage 1 (or an anchored variant—this choice does not affect the argument as long as it is cross-fitted and converges to a well-defined limit). Define the plug-in anchored CATE at covariate value x by

$$\widehat{\tau}^{\text{anch}}(x) := \widehat{\mu}^{\text{anch}}_{1,0}(x) - \widehat{\mu}^{\text{anch}}_{0,0}(x).$$

b) Orthogonal pseudo-outcome.: For each observation i from site c_i, define the DR pseudo-outcome

$$\psi_i := \widehat{\tau}^{\text{anch}}(X_i) + \frac{A_i - e_{c_i}(X_i)}{e_{c_i}(X_i)\{1 - e_{c_i}(X_i)\}} \Big\{ Y_i - \widehat{\mu}^{\text{anch}}_{A_i, c_i}(X_i) \Big\}. \tag{5}$$

We estimate $\tau^\star(\cdot)$ by regressing ψ_i on X_i using any learner $\mathcal{T}$ with cross-fitting:

$$\widehat{\tau}(x) := \mathcal{T}\big(\{(X_i, \psi_i)\}_{i=1}^{N}\big)(x),$$

where the nuisance functions inside ψ_i are trained on folds excluding the observation being evaluated. For a pointwise statement, one may view $\mathcal{T}$ as a local smoother or series estimator; for notational simplicity we present the argument as a conditional-mean functional with a localizer $K_h(\cdot)$ (below), which yields a standard influence-function proof.

C. Assumptions used in this appendix

We rely on Assumptions **A1**–**A6** stated in the main text. For the asymptotic argument, we also use the following probabilistic regularity conditions, which are independent of Assumptions **A4**–**A6**.

a) Triangular-array CLT conditions.: We assume:

- (P1) **Independence of sampling.** Conditional on S, $\{O_{ci}\}_{i=1}^{n_c}$ are independent within each site and independent across sites.
- (P2) **Finite second moments.** For each fixed x, the influence contributions defined below have finite second moments.
- (P3) **Stabilized site proportions.** C is fixed and $\pi_c \to \pi^\star_c \in [0, 1]$ with $\sum_{c=0}^{C} \pi^\star_c = 1$.

Assumptions (P1)–(P3) are standard probabilistic conditions required to establish asymptotic normality for estimators constructed from heterogeneous samples and are conceptually distinct from the causal assumptions used to define the estimand.

Assumption (P1) formalizes independent sampling within and across trials. In the RCT settings motivating this work, trial participants are typically enrolled independently within each site, and different trials are conducted independently under separate protocols. This assumption rules out interference or cross-trial dependence but does not impose any restriction on treatment effects or covariate distributions across sites. Assumption (P2) requires finite second moments of the influence-function contributions. This is a mild condition in randomized trials with bounded or sub-Gaussian outcomes and known treatment propensities bounded away from 0 and 1. It ensures that no single observation dominates the estimator and is standard in influence-function-based analyses. Assumption (P3) specifies a triangular-array asymptotic regime in which the number of sites is fixed and site-level sample proportions stabilize as the total sample size grows. This reflects common multi-trial meta-analytic settings, where a fixed collection of trials contributes increasing numbers of participants, and ensures that asymptotic variance reflects a stable mixture of site-specific contributions rather than being dominated by a vanishing or exploding subset of sites.

Importantly, they are do not impose additional identification or transportability assumptions beyond those stated in Assumptions **A1**–**A6**.

b) Cross-fitting.: All nuisance estimators used to construct ψ_i are evaluated out-of-fold so that, conditional on the training sample, evaluation observations are independent of the fitted nuisances.

D. Working-model limits and approximation terms

To make the dependence on **A5**–**A6** explicit, we define the *working-model* limits and the associated structural approximation errors.

*a) Proxy limits and cross-site deviations (**A5**).:* Let μ_a^{proxy} denote the population limit of the Stage 1 proxy learner trained on source sites. Assumption **A5** posits that for each site c and arm a,

$$\begin{aligned}\mu_{a,c}(x) &= \mu_a^{\text{proxy}}(x) + \delta_{a,c}(x),\\ \delta_{a,c} &\in \mathcal{D},\end{aligned}$$

where $\mathcal{D}$ is a restricted low-complexity class (e.g., approximately sparse linear corrections). When **A5** holds only approximately, define the within-arm approximation error

$$r_{a,c}(x) := \mu_{a,c}(x) - \mu_a^{\text{proxy}}(x) - \Pi_{\mathcal{D}}\{\mu_{a,c} - \mu_a^{\text{proxy}}\}(x),$$

where $\Pi_{\mathcal{D}}$ denotes the best approximation (projection) onto $\mathcal{D}$ in $L_2(P_0)$. For the target placebo arm $(a,c) = (0,0)$, Stage 2 estimates an element of $\mathcal{D}$ from gold placebo data; the residual $r_{0,0}$ captures structural misspecification of **A5** for that arm.

*b) Cross-arm coupling (**A6**).:* When treated target outcomes are unavailable, Assumption **A6** links the treated-arm bias to the placebo-arm bias:

$$\delta_{1,0}(x) = \rho\, \delta_{0,0}(x) + \zeta(x), \qquad \rho \in [0,1].$$

This defines the working-model treated regression $\mu_{1,0}^\star$ (and hence $\tau^\star$) in disconnected regimes. Under Option B in the main text we adopt the specialization $\rho = 1$ and $\zeta \equiv 0$ as a pragmatic working choice. In general, the irreducible cross-arm nontransfer component ζ contributes a structural approximation term that does not vanish with sample size.

E. A score-based representation for pointwise inference

We derive a pointwise asymptotic linear representation for the working-model CATE $\tau^\star(x)$ without introducing kernel or localization arguments. Throughout, x is treated as fixed.

Recall that Stage 3 constructs a doubly robust pseudo-outcome

$$\psi_i = \frac{A_i - e_{S_i}}{e_{S_i}(1 - e_{S_i})}\Big\{Y_i - \mu^\star_{A_i,S_i}(X_i)\Big\} + \tau^\star(X_i), \qquad (6)$$

where e_{S_i} is the known site-specific propensity and $\mu^\star_{a,s}$ denotes the population limits of the anchored outcome regressions under Assumptions **A4**–**A6**.

The Stage 3 estimator $\hat{\tau}(\cdot)$ is obtained by regressing ψ_i on X_i using cross-fitting and a sufficiently rich regression class. For any fixed covariate value x, the target parameter is

$$\tau^\star(x) = \mathbb{E}[\psi_i \mid X_i = x]\,.$$

a) Oracle score representation.: Define the oracle score [37]

$$\phi(O;x) := \frac{A - e_S}{e_S(1 - e_S)}\Big\{Y - \mu^\star_{A,S}(X)\Big\} + \tau^\star(X) - \tau^\star(x). \qquad (7)$$

By construction,

$$\mathbb{E}[\phi(O;x) \mid X = x] = 0, \qquad \mathbb{E}[\phi(O;x)] = 0,$$

and $\phi(O;x)$ has finite second moments under Assumption **A4** and bounded outcomes.
The oracle score $\phi(O;x)$ is a random variable indexed by a fixed covariate value x. Here, the capital letter X denotes the random covariate contained in the observation $O = (X, A, Y, S)$, while the lowercase x denotes the specific covariate value at which the CATE $\tau^\star(x)$ is evaluated. The score is constructed so that, conditional on $X = x$, its expectation is zero: $\mathbb{E}[\phi(O;x) \mid X = x] = 0$. Intuitively, $\phi(O;x)$ measures how much a single observation contributes to estimating the treatment effect *at the covariate value* x, rather than at its own realized covariate X. This distinction is standard in pointwise influence-function representations for regression functionals.

b) Asymptotic linear expansion.: Under Neyman orthogonality of the score in (6), cross-fitting, and the nuisance rate conditions stated, standard arguments for orthogonal regression functionals yield the expansion

$$\hat{\tau}(x) - \tau^\star(x) = \frac{1}{N}\sum_{i=1}^{N} \phi(O_i;x) + R_N(x), \qquad (8)$$

where the remainder satisfies $R_N(x) = o_p(N^{-1/2})$.

Importantly, this expansion holds pointwise in x without requiring kernel localization or bandwidth selection. The influence function $\phi(O;x)$ captures the first-order sensitivity of the estimator at the fixed covariate value x.

c) Asymptotic normality.: Because observations are independent but not identically distributed across sites, the summands in (8) form a triangular array. Under finite second moments and stabilization of site proportions, a Lindeberg–Feller central limit theorem implies

$$\sqrt{N}\big(\widehat{\tau}(x) - \tau^\star(x)\big) \xrightarrow{\mathcal{D}} \mathcal{N}\left(0,\ \sum_{c=0}^{C} \pi_c^\star \operatorname{Var}\big(\phi_c(O_{c1};x)\big)\right).$$

This result establishes valid $\sqrt{N}$-rate pointwise inference for the working-model CATE $\tau^\star(x)$. No claim is made regarding nonparametric identification or semiparametric efficiency for the true target CATE $\tau_0(x)$.

F. Main theorem

Theorem 1 (Asymptotic linearity and $\sqrt{N}$-normality for the working-model CATE)**.** *Fix a covariate value x. Under (P1)–(P3), cross-fitting, Assumptions **A1–A6**, and the rate conditions in* (11)*, the Stage 3 estimator admits the pointwise expansion*

$$\widehat{\tau}(x) - \tau^\star(x) = \frac{1}{N}\sum_{c=0}^{C}\sum_{i=1}^{n_c}\phi_c(O_{ci};x) + R_N(x), \tag{9}$$

$$R_N(x) = o_p(N^{-1/2}) + \mathrm{Approx}(x), \tag{10}$$

where $\phi_c(\cdot;x)$ is a site-specific influence function satisfying $\mathbb{E}[\phi_c(O_{c1};x) \mid S=c] = 0$ and $\mathbb{E}[\phi_c(O_{c1};x)^2] < \infty$.

If $\mathrm{Approx}(x) = 0$ (exact working model), then

$$\sqrt{N}\big(\widehat{\tau}(x) - \tau^\star(x)\big) \xrightarrow{\mathcal{D}} \mathcal{N}\left(0,\ \sum_{c=0}^{C}\pi_c^\star \operatorname{Var}\big(\phi_c(O_{c1};x)\big)\right).$$

No claim of semiparametric efficiency for the true target CATE $\tau_0(x)$ is made.

a) Rate conditions.: Let $n_{\mathrm{proxy}} := \sum_{c=1}^{C} n_c$ be the proxy (source) sample size and let

$$m_0 := \big|\{i : S_i = 0,\ A_i = 0\}\big|$$
$$\text{(and } m_1 := |\{i : S_i = 0,\ A_i = 1\}| \text{ if Option A is used).}$$

Define the proxy prediction error on the target covariate distribution

$$\delta_{\mathrm{proxy}} := \big\|\widehat{\mu}_0^{\mathrm{proxy}} - \mu_0^{\mathrm{proxy}}\big\|_{L_2(P_0)} + \big\|\widehat{\mu}_1^{\mathrm{proxy}} - \mu_1^{\mathrm{proxy}}\big\|_{L_2(P_0)},$$

and the gold-calibration error

$$\delta_{\mathrm{gold}} := \big\|\widehat{\delta}_{0,0} - \delta_{0,0}^{\dagger}\big\|_{L_2(P_0)},$$

where $\delta_{0,0}^{\dagger}$ denotes the best-approximation element in $\mathcal{D}$ for the target placebo deviation. Assume

$$\begin{aligned} \delta_{\mathrm{proxy}}\,\delta_{\mathrm{gold}} &= o_p(N^{-1/2}),\\ \delta_{\mathrm{proxy}} &= o_p(1),\\ \delta_{\mathrm{gold}} &= o_p(1). \end{aligned} \tag{11}$$

Note that the gold sample sizes m_0 (and m_1 if Option A is used) enter the analysis only through the nuisance rate δ_{gold} and therefore affect the second-order remainder term, but do not influence the $\sqrt{N}$ oracle fluctuation or asymptotic variance.

G. Proof of Theorem 1

Proof. We decompose the error into an oracle empirical-process term plus a nuisance plug-in term.

a) Step 1: Plug-in decomposition into oracle and estimation terms.: Write

$$\widehat{\tau}(x) - \tau^\star(x) = \underbrace{\frac{1}{N}\sum_{i=1}^{N}\Big(\widehat{\psi}_i(x) - \psi_i^\star(x)\Big)}_{\text{(I) (plug-in / nuisance error)}} + \underbrace{\frac{1}{N}\sum_{i=1}^{N}\psi_i^\star(x) - \mathbb{E}[\psi^\star(O;x)]}_{\text{(II) (oracle fluctuation)}},$$

where $\psi_i^\star(x) := \psi^\star(O_i;x)$ denotes the oracle score and $\widehat{\psi}_i(x)$ is its cross-fitted plug-in analogue. This is the standard empirical process decomposition [38].

b) Step 2: Control of plug-in term (I) via orthogonality and cross-fitting.: By cross-fitting, conditional on the training sample used to fit nuisances, the evaluation observations are independent of the fitted nuisances. A standard second-order expansion of the map $\mu \mapsto \mathbb{E}[\psi^\star(O;x)]$ around $\mu^\star$ together with Neyman orthogonality implies that the leading contribution of nuisance error is second order [35]. Consequently,

$$\text{(I)} = O_p\big(\delta_{\mathrm{proxy}}\,\delta_{\mathrm{gold}}\big) + \mathrm{Approx}(x),$$

where the approximation term $\mathrm{Approx}(x)$ collects structural misspecification from Assumptions A5–A6:

$$\begin{aligned} \mathrm{Approx}(x) := {} & |r_{0,0}(x)| \\ & + |\zeta(x)| \\ & + |r_{1,0}(x)|\mathbb{I}\{\text{ if Option A used}\}. \end{aligned}$$

Assumption **A4** is used to control the evaluation of proxy errors on the target covariate distribution (i.e., to ensure the $L_2(P_0)$ errors above are well-defined and do not blow up under moderate covariate shift). Assumptions **A5–A6** define the working-model limits and ensure that the gold correction targets a stable low-complexity object; violations appear as the explicit $\mathrm{Approx}(x)$ term.

Under (11), $\delta_{\mathrm{proxy}}\,\delta_{\mathrm{gold}} = o_p(N^{-1/2})$, hence

$$\text{(I)} = o_p(N^{-1/2}) + \mathrm{Approx}(x).$$

c) Step 3: Triangular-array CLT for oracle term (II).: Define the centered contributions

$$\phi_c(O_{ci};x) := \psi^\star(O_{ci};x) - \mathbb{E}\big[\psi^\star(O;x) \mid S=c\big],$$

so that

$$\text{(II)} = \frac{1}{N}\sum_{c=0}^{C}\sum_{i=1}^{n_c}\phi_c(O_{ci};x).$$

By (P1) the summands are independent; by (P2) they have finite variance; and by (P3) the site proportions stabilize. A triangular-array (Lindeberg–Feller) CLT yields

$$\sqrt{N}\,\text{(II)} \Rightarrow \mathcal{N}\left(0,\ \sum_{c=0}^{C}\pi_c^\star \operatorname{Var}\big(\phi_c(O_{c1};x)\big)\right).$$

d) Step 4: Combine.: Combining Steps 2–3 gives the expansion (9)–(10). When $\mathrm{Approx}(x) = 0$, $\sqrt{N}(\mathrm{I}) = o_p(1)$ and the stated asymptotic normality follows. $\square$

H. Finite-sample error decomposition and sample-size dependence

The asymptotic linear representation derived above yields a transparent finite-sample decomposition that clarifies how the different data regimes enter the estimation error and how orthogonality governs the interaction between proxy and gold samples.

a) Pointwise decomposition.: For each fixed covariate value x, the estimation error relative to the working-model CATE admits the bound

$$|\widehat{\tau}(x) - \tau^\star(x)| \le \underbrace{\Big|\frac{1}{N}\sum_{c=0}^{C}\sum_{i=1}^{n_c}\phi_c(O_{ci};x)\Big|}_{\text{oracle sampling fluctuation}} + \underbrace{C_1\,\delta_{\text{proxy}}\,\delta_{\text{gold}}}_{\text{second-order nuisance remainder}} \quad (12)$$
$$+ \underbrace{\mathrm{Approx}(x)}_{\textbf{A5}\text{–}\textbf{A6}\text{ structural approximation bias}}, \quad (13)$$

where $\phi_c(\cdot;x)$ is the site-specific influence function defined in Theorem 1, δ_{proxy} denotes the prediction error of the Stage 1 proxy learner evaluated on the target covariate distribution, and δ_{gold} denotes the estimation error of the Stage 2 gold calibration. The first term is an oracle empirical-process fluctuation of order $O_p(N^{-1/2})$. The second term arises from plug-in estimation of nuisance components and is second order due to Neyman orthogonality and cross-fitting. The final term collects approximation errors induced by violations of the structural regularity conditions in Assumptions **A5–A6** and does not vanish with increasing sample size.

Taking $L_2(P_0)$ norms yields the corresponding global bound

$$\|\widehat{\tau} - \tau^\star\|_{L_2(P_0)} \lesssim O_p(N^{-1/2}) + \delta_{\text{proxy}}\,\delta_{\text{gold}} + \|\mathrm{Approx}\|_{L_2(P_0)}. \quad (14)$$

b) Proxy sample size dependence.: Let $n_{\text{proxy}} = \sum_{c=1}^{C} n_c$ denote the total sample size of the source trials used to fit the Stage 1 proxy outcome models. Under standard learning-theoretic guarantees for regression estimators [38],

$$\delta_{\text{proxy}} = O_p\big(a(n_{\text{proxy}})\big),$$

where the rate function $a(\cdot)$ depends on the complexity of the proxy learner (e.g., $a(n) \asymp n^{-1/2}$ for parametric models, or slower rates for nonparametric and machine learning estimators). This term reflects the accuracy with which outcome regressions learned on abundant source data generalize to the target covariate distribution.

c) Gold placebo sample size dependence.: Let m_0 denote the number of placebo observations available in the target site, which are used to estimate the Stage 2 transport-bias correction in the restricted class $\mathcal{D}$. In the canonical sparse linear setting with ambient dimension p and sparsity level s, standard high-dimensional theory yields

$$\delta_{\text{gold}} = O_p\Big(\sqrt{\tfrac{s\log p}{m_0}}\Big)$$

under appropriate design conditions such as restricted eigenvalue or compatibility assumptions [39, 33]. This term characterizes the *small-gold regime*: even when proxy data are abundant, uncertainty in the calibration step can dominate finite-sample behavior when m_0 is limited.

d) Combined scaling.: Substituting the typical rates into (14) yields

$$\|\widehat{\tau} - \tau^\star\|_{L_2(P_0)} \lesssim O_p(N^{-1/2}) + O_p\Big(a(n_{\text{proxy}})\sqrt{\tfrac{s\log p}{m_0}}\Big) + \|\mathrm{Approx}\|_{L_2(P_0)}. \quad (15)$$

Equation (15) makes explicit that: (i) the $\sqrt{N}$ sampling term depends on the total Stage 3 sample size $N = \sum_{c=0}^{C} n_c$, (ii) the nuisance-driven remainder depends jointly on the proxy and gold sample sizes through an orthogonality-induced product structure, and (iii) violations of Assumptions **A5–A6** contribute an explicit approximation bias that does not vanish asymptotically.

e) Relation to the true target CATE and efficiency.: Finally, the total error relative to the true target-site CATE decomposes as

$$|\widehat{\tau}(x) - \tau_0(x)| \le |\widehat{\tau}(x) - \tau^\star(x)| + |\tau^\star(x) - \tau_0(x)|.$$

The first term is controlled by the stochastic and second-order components in (13). The second term represents a structural bias induced by the working-model restrictions in Assumptions **A5–A6** (e.g., nonzero cross-arm nontransfer components or approximation residuals) and need not vanish in disconnected regimes. Because $\tau_0(x)$ is not nonparametrically identified without additional assumptions and Assumptions **A5–A6** define a restricted approximation class rather than full transportability, we do *not* claim semiparametric efficiency for $\tau_0(x)$. Instead, orthogonality yields valid $\sqrt{N}$-rate pointwise inference for the working-model target $\tau^\star(x)$ under the stated conditions.